

DIRECT METHODS FOR LINEAR SYSTEMS

$$Ax = b \quad A \in \mathbb{R}^{n \times n} \quad b \in \mathbb{R}^n \quad A \text{ invertible}$$

Direct methods $\leadsto$ they use matrix factorizations

Example: $A = LU = \begin{matrix} \square & \square \end{matrix} \quad L \tilde{U} x = b$

$$\begin{cases} Ly = b \\ Ux = y \end{cases} \quad \text{these are triangular lin. sys} \rightarrow O(n^2)$$

$$A = QR = \begin{matrix} \square & \square \end{matrix} \quad \begin{matrix} \text{orthogonal} \\ \downarrow \\ O(n^3) \end{matrix}$$

$$QRx = b \Leftrightarrow Rx = Q^T b \rightarrow O(n^2)$$

If A is symmetric: let $A = Q D Q^T$ be a spectral decomp. of A .

$$A = \begin{matrix} \square & \square & \square \end{matrix} \quad \begin{matrix} \text{orth} & \lambda_1 \dots \lambda_n & \text{orth}^T \end{matrix}$$

$$Q D Q^T x = b \Leftrightarrow D Q^T x = Q^T b$$

$$\Leftrightarrow Q^T x = D^{-1} Q^T b$$

$\hookrightarrow$ computed in $O(n^2)$

$\hookrightarrow O(n)$: first entry divided by λ_1 , ..., last entry divided by λ_n

$$\Leftrightarrow x = Q \cdot (D^{-1} Q^T b)$$

$\hookrightarrow$ another $O(n^2)$ operations

if A is symmetric $A = L D L^T = \begin{matrix} \square & \square & \square \end{matrix} \quad \begin{matrix} d_1 & 0 \\ 0 & d_n \end{matrix}$

$$\underbrace{L D L^T}_y x = b$$

$$Ly = b \quad O(n^2)$$

$$Dz = y \quad O(n)$$

$$L^T x = z \quad O(n^2)$$

once we have a factorization of A where the factors are

orthogonal, diagonal, and/or triangular, we can solve $Ax = b$ in $O(n^2)$ operations.

LU factorization - Gaussian elimination

$$A_1 = A = \begin{bmatrix} \textcircled{2} & 1 & -1 \\ 1 & 0 & 2 \\ -4 & 3 & 2 \end{bmatrix}$$

$\longrightarrow$

$$\begin{bmatrix} 2 & 1 & -1 \\ 0 & -1/2 & 5/2 \\ 0 & 5 & 0 \end{bmatrix} =: A_2 = E_1 A_1$$

$$E_1 = \begin{bmatrix} 1 & 0 & 0 \\ -1/2 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} \rightarrow \text{elementary Gauss matrix}$$

$\hookrightarrow$ multiplying factors that we needed to introduce the zeros in the first column.

$$A_2 = \begin{bmatrix} 2 & 1 & -1 \\ 0 & -1/2 & 5/2 \\ 0 & 5 & 0 \end{bmatrix}$$

$$A_3 = \begin{bmatrix} 2 & 1 & -1 \\ 0 & -1/2 & 5/2 \\ 0 & 0 & 25 \end{bmatrix} = E_2 \cdot A_2$$

$$E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 10 & 1 \end{bmatrix}$$

$$U = E_2 A_2 = E_2 E_1 A_1 = E_2 E_1 A \rightsquigarrow E_2^{-1} U = E_1 A$$

$$\leadsto A = \underbrace{E_1^{-1} E_2^{-1}} U$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 1/2 & 1 & 0 \\ -2 & -10 & 1 \end{bmatrix} =: L$$

In general :

$$A_1 = A = \begin{bmatrix} * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \end{bmatrix} \xrightarrow{F_1} \begin{bmatrix} 1 & & & \\ -L_{21} & 1 & & \\ -L_{31} & & \ddots & \\ \vdots & & & 1 \end{bmatrix}$$

$$A_2 = E_1 A_1 = \begin{bmatrix} * & & \\ 0 & & \\ 0 & & \\ 0 & & \end{bmatrix} \begin{bmatrix} * & & \\ & & \\ & & \\ & & \end{bmatrix}$$

$$L_{j1} = \frac{A_{j1}}{A_{11}}$$

At step k : $A_k =$

$$E_k = \begin{bmatrix} 1 & & & & \\ & \ddots & & & \\ & & 1 & & \\ & & & \ddots & \\ & & & & 1 \end{bmatrix}$$

$$L_{j,k} := \frac{(A_k)_{jk}}{(A_k)_{kk}}$$

$$A_n = U = E_{n-1} E_{n-2} \dots E_1 A$$

$$\leadsto A = (E_1^{-1} E_2^{-1} \dots E_{n-1}^{-1}) \cdot U$$

$$L = \begin{bmatrix} 1 & & & \\ & L_{21} & & 0 \\ & L_{31} & L_{32} & \\ \vdots & \vdots & \vdots & \ddots & 1 \end{bmatrix}$$

cost: $O(n^2 + (n-1)^2 + (n-2)^2 + \dots + 4^2 + 3^2 + 2^2 + 1^2)$
 $= O(n^3) \rightsquigarrow \approx \frac{2}{3}n^3$ operations
($\frac{4}{3}n^3$ operations for QR)

Note: if $A = \begin{bmatrix} 0 & 1 & 2 \\ 3 & 4 & 5 \\ 6 & 7 & 8 \end{bmatrix} \rightarrow$ the first step cannot be done!

$$\text{if } A = \begin{bmatrix} 10^{-10} & 1 & 2 \\ 1 & 4 & 5 \\ 6 & 7 & 8 \end{bmatrix} \longrightarrow \begin{bmatrix} 10^{-10} & 1 & 2 \\ 0 & 4-10^{-10} & 5-2 \cdot 10^{10} \\ \dots & \dots & \dots \end{bmatrix}$$

There is a stability problem!

One can show that the computed $\tilde{L}, \tilde{U}$ are the exact LU factorization of $\hat{A} := \tilde{L} \tilde{U}$ s.t.

$$\|A - \hat{A}\| \leq \underbrace{\|\tilde{L}\| \cdot \|\tilde{U}\|}_{\text{machine precision}} \cdot O(n)$$

$\hookrightarrow$ this may be much larger than $\|A\|$!!

This problem is connected to having small pivots!

Fix: PIVOTING! At each step, we choose the row with largest pivot, and we swap it with the k -th row, and then do the step of Gaussian elimination.

Example (revisited):

$$A_1 = A = \begin{bmatrix} 2 & 1 & -1 \\ 1 & 0 & 2 \\ -4 & 3 & 2 \end{bmatrix} \xrightarrow{\text{1st step Gaussian elimination}} \begin{bmatrix} -4 & 3 & 2 \\ 1 & 0 & 2 \\ 2 & 1 & -1 \end{bmatrix} \xrightarrow{\text{1st step Gaussian elimination}} \begin{bmatrix} -4 & 3 & 2 \\ 0 & 3/4 & 5/2 \\ 0 & 5/2 & 0 \end{bmatrix} = A_2$$

$$A_2 = \begin{bmatrix} -4 & 3 & 2 \\ 0 & 3/4 & 5/2 \\ 0 & 5/2 & 0 \end{bmatrix} \xrightarrow{\text{2nd step Gaussian elim.}} \begin{bmatrix} -4 & 3 & 2 \\ 0 & 5/2 & 0 \\ 0 & 3/4 & 5/2 \end{bmatrix} \xrightarrow{\text{2nd step Gaussian elim.}} \begin{bmatrix} -4 & 3 & 2 \\ 0 & 5/2 & 0 \\ 0 & 0 & 5/2 \end{bmatrix} = U$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ -1/2 & 1 & 0 \\ -1/4 & 3/10 & 1 \end{bmatrix}$$

By design, the entries of L below the diagonal have magnitude ≤ 1

⚠ $LU \neq A$ now $\rightsquigarrow LU = P \cdot A$

$\hookrightarrow P$ permutation matrix

$n \times n$ matrix with exactly one "1" in each column and in each row

In the example: $\begin{bmatrix} \text{1st row} \\ \text{2nd row} \\ \text{3rd row} \end{bmatrix} \rightsquigarrow \begin{bmatrix} \text{3rd row} \\ \text{2nd row} \\ \text{1st row} \end{bmatrix} \rightsquigarrow \begin{bmatrix} \text{3rd row} \\ \text{1st row} \\ \text{2nd row} \end{bmatrix}$

It's as if we did "standard" (non-pivoted) Gaussian elimination

$$\text{on } \tilde{A} = \begin{bmatrix} -4 & 3 & 2 \\ 2 & 1 & -1 \\ 1 & 0 & 2 \end{bmatrix} \rightarrow \tilde{A} = \underset{\substack{= \\ P \cdot A}}{L} U$$

Ex: Find P s.t. $P \cdot A = \tilde{A}$

$$\hookrightarrow P = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

With pivoting we obtain a pivoted LU factorization:

$$P \cdot A = L \cdot U$$

Permutation matrices are orthogonal $\leadsto A = P^T L U$

$$Ax = b$$

$$P^T L U x = b$$

$$L U x = P b$$

Usually stable, except for a small set of matrices.

Now let us assume that $A \in \mathbb{R}^{n \times n}$ symmetric (and invertible)

$$A = \boxed{} \longrightarrow \boxed{\begin{smallmatrix} * \\ 0 \\ 0 \\ 0 \end{smallmatrix}}$$

$$\begin{bmatrix} 2 & 1 \\ 1 & 8 \end{bmatrix} \longrightarrow \begin{bmatrix} 2 & 1 \\ 0 & * \end{bmatrix}$$

Standard G.E. does NOT preserve symmetry $\ddot{}$

$$A_1 = A$$

$$A_2 = E_1 \cdot A_1 \cdot E_1^T$$

$\leadsto E_j$ defined as before

$$\vdots$$

$$A_k = E_{k-1} A_{k-1} E_{k-1}^T$$

$$\vdots$$

$$\begin{bmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{bmatrix}$$

$$\longrightarrow$$

$$A_2 = E_1 A_1 E_1^T =$$

$$\begin{bmatrix} * & 0 & 0 & 0 \\ 0 & * & * & * \\ 0 & * & * & * \\ 0 & * & * & * \end{bmatrix}$$

$$A_k = \begin{bmatrix} \overset{k}{*} & \overset{k}{0} & \overset{k}{0} & \overset{k}{0} \\ \overset{k}{0} & \overset{k}{*} & \overset{k}{*} & \overset{k}{*} \\ 0 & \text{[scribble]} \end{bmatrix}$$

$A_n \leadsto$ diagonal matrix $=: D$

$$D = A_n = E_{n-1} \underbrace{A_{n-1}} E_{n-1}^T = E_{n-1} E_{n-2} A_{n-2} E_{n-2}^T E_{n-1}^T = \dots$$

$$D = E_{n-1} E_{n-2} \dots E_1 A E_1^T E_2^T \dots E_{n-1}^T$$

$$\Rightarrow A = \underbrace{(E_1^{-1} E_2^{-1} \dots E_{n-1}^{-1})}_L \cdot D \cdot \underbrace{(E_{n-1}^{-T} E_{n-2}^{-T} \dots E_1^{-T})}_{=L^T}$$

$$A = L D L^T = \begin{array}{|c|} \hline \text{lower triangular} \\ \hline \end{array} \begin{array}{|c|} \hline \text{diagonal} \\ \hline \end{array} \begin{array}{|c|} \hline \text{upper triangular} \\ \hline \end{array} \quad \text{"LDL decomposition"}$$

↳ ⚠ This is NOT a spectral decomposition!

We can save some time during the Gaussian Elimination:

we only need to complete the lower triangular part of

$A_1, A_2, \dots, A_n$ and the rest follows by symmetry

$$\leadsto \text{cost} \approx \frac{1}{3}n^3.$$

Pivoting $\leadsto$ a bit more complicated, for stability reasons
D should have some diagonal blocks $2 \times 2 \dots$

If $A \in \mathbb{R}^{n \times n}$ symmetric positive definite

The good news is that the pivots d_{kk} encountered during the Gaussian elimination process are all > 0 .

$$A = L D L^T = L D^{1/2} \cdot D^{1/2} L^T = \underbrace{(L D^{1/2})}_{R^T} \cdot \underbrace{(L D^{1/2})^T}_R = R^T R$$

$\downarrow$
 $\begin{bmatrix} d_{11} & & 0 \\ & d_{22} & \\ 0 & & d_{nn} \end{bmatrix} \rightarrow D^{1/2} = \begin{bmatrix} \sqrt{d_{11}} & & 0 \\ & \ddots & \\ 0 & & \sqrt{d_{nn}} \end{bmatrix}$

R
↳ upper triangular

Def: $A = R^T R$, R upper triangular is a Cholesky decomposition

$Ax = b$
How do I solve it?

is A small?

Yes

Direct method

- LU if A non-sym
- LDL if A sym
- Cholesky if A SPD

$$O(n^3)$$

No

Iterative methods

- GNRES if A is non-sym
- MINRES if A is symmetric
- Conjugate gradient if A is SPD

$$O\left(n \cdot (\#iter)^2 + \#iter \cdot \left(\begin{array}{l} \text{cost of one} \\ \text{matrix-vector} \\ \text{mult. with A} \end{array}\right)\right)$$

exception to the rule:

large SPARSE matrices for which we (somehow) can get sparse L and sparse U factors.